

<https://github.com/surabhi84ard/Mini/tree>

Abstract

This manual shows how to use the 7474 D-Flip Flop ICs in a sequential circuit to realize a decade counter.

1 Components

Component	Value	Quantity
Breadboard	–	1
Resistor	$\geq 220\Omega$	1
Vaman	1	
Seven Segment Display	Common Anode	1
Decoder	7447	1
Flip Flop	7474	2
Jumper Wires	–	20

Table 1: Required Components

2 7474 Pin Diagram

Pin	Function
14	V _{cc}
13	CLK2
12	D2
11	Q2
10	—
9	—
8	—
7	—
6	—
5	CLK1
4	D1
3	Q1
2	—
1	GND

3 Decade Counter Setup

3.1 Vaman to Flip-Flop and Decoder Connections

Vaman Pin	Connected To
D6	CLOCK
D7	W
D8	X
D9	Y
D2	Z
D3	A
D4	B
D5	C
D13	D

Table 2: Vaman to Logic Pins

3.2 Decoder Connections

4 Logic Flow

The 7474 flip-flops are used to generate a 4-bit binary counter. The output is incremented on each clock pulse and decoded by the 7447 to drive the seven segment display.

Decoder Pin	Connected To
7	W
1	X
2	Y
6	Z
16	V _{cc}

Table 3: 7447 Decoder Pin Mapping

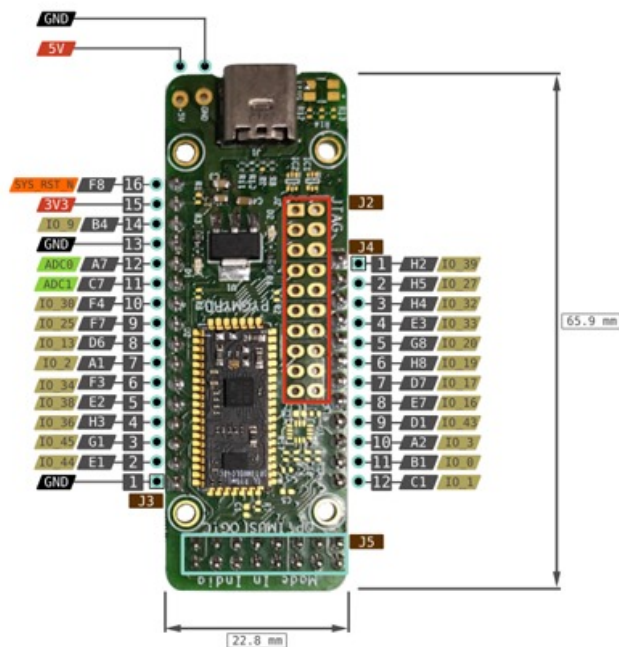
4.1 Truth Table

CLOCK	W	X	Y	Z	A	B	C	D
0	0	0	0	0	0	0	0	1
1	0	0	0	1	0	0	1	0
2	0	0	1	0	0	1	1	0
3	0	0	1	1	1	0	0	0
4	0	1	0	0	1	0	1	0
5	0	1	0	1	1	1	0	0
6	0	1	1	0	0	0	1	1
7	0	1	1	1	0	1	0	0
8	1	0	0	0	1	0	0	0
9	1	0	0	1	0	0	0	0

Table 4: Decade Counter Output

5 Figures

PYGMY BB v1 PINOUT



- Reset
- Power
- GND
- IO/Pad Number
- Physical Pin/Ball
- Analog Input
- Pin Function(s)
- Component Pin
- Board Header Pin

On-Board Components

SPI FLASH Memory (on Pygmy Stamp)

SS	IO_39 H2	SPI_MASTER_SSn1
SCLK	IO_34 F3	SPI_MASTER_CLK
SI	IO_38 E2	SPI_MASTER_MOSI
SO	IO_38 H3	SPI_MASTER_MISO

Buttons

USR	IO_6 B3	GPIO[0]
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RGB LED

RED	IO_22 G7	GPIO[6]
GREEN	IO_27 H7	GPIO[5]
BLUE	IO_16 E8	GPIO[4]

BM1160 ACCEL + GYRO

SCx	IO_0 B1	SCL_0
SDx	IO_1 C1	SDA_0

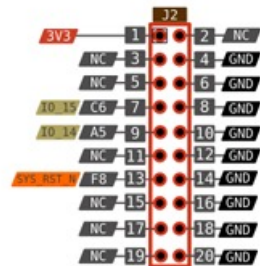
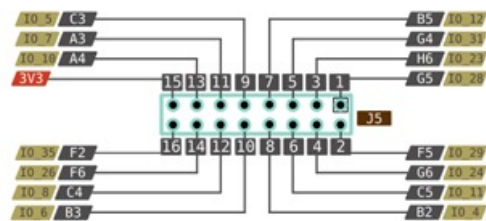


Figure 1: Pygmy BB v1 Pin Diagram